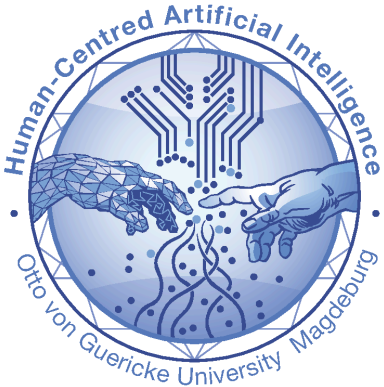


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Explainable Medical Question Answering using Multi-Agent RAG Systems

Human Centred Natural Language Processing

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Agenda

- Introduction
- Related Work
- Chunking Methods
- Experiments
- Results
- Summary



Introduction

Limitations of Standard Approaches

- Medical QA requires complex reasoning over specialised domain knowledge
- Standard systems predict answers directly from given options, which would be unrealistic in clinical settings
- In the real world, answer options are rarely available thus models must demonstrate genuine reasoning
- This highlights the need for systems that are both **accurate** and **explainable**

Our Approach

- Combine **RAG** with a **multi-agent architecture** for explainable medical QA
- Evaluate on **MedQA** using 1,273 USMLE questions with full textbook retrieval corpus
- Test both **with-options** and **without-options** inference conditions
- Measure **faithfulness** and not just accuracy

Key Contributions:

Chunking Analysis | **Multi-Agent vs Single** | **Options vs No Options** | **Faithfulness Evaluation**

Why can't we just give the model the answer options and pick the best one?

- **The Real-World Gap:**
  - In clinical practice, a doctor reasoning through a case has no pre-listed answer options.
  - Standard MCQ benchmarks create a false sense of model capability by making options available at inference.
- **Pattern Matching Risk:**
  - Models given options can exploit option-level surface cues to guess the answer without genuine reasoning.
  - Removing options forces the model to actually understand and reason through the question.
- **Explainability Deficit:**
  - Most RAG systems return an answer with retrieved passages but no structured reasoning.
  - Clinicians need to understand WHY a diagnosis is supported and not just what the top answer letter is.
- **Our Solution:**
  - Multi-agent RAG system retrieves domain evidence, generates pros/cons reasoning per option, and evaluates both with and without options & measuring faithfulness, not just accuracy.

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Related Work

Related Work

- Early models like BioBERT, PubMedBERT showed strong performance
- However, they mainly learn patterns from data
- Limited ability to perform step-by-step reasoning

Recent Advances

- RAG & Multi-Agent Systems

- RAG improves answers by using external medical knowledge
- Some approaches use structured reasoning techniques
- Multi-agent systems:
  - Break tasks into smaller steps
  - Improve reasoning process

Related Work

Retrieval Methods

- **Adaptive-RAG:**
  - Routes queries to single-step or multi-step retrieval based on question complexity
- **i-MedRAG:**
  - Iterative follow-up subqueries, each shaped by previous retrievals
- **RAG2:**
  - Uses model's own reasoning trace as retrieval query, then filters chunks

Reasoning & Structure

- **MedCoT-RAG:**
  - Structures output around clinical stages such as symptoms, underlying mechanisms and differential diagnosis generation
- **PubMedBERT / BioBERT:**
  - Domain fine-tuned transformers which are strong on MCQ but rely on surface patterns, not genuine reasoning
- **Our System:**
  - Distributes clinical reasoning across specialised agents; adds faithfulness evaluation beyond accuracy

Related Work

Chunking Strategies

- **RAPTOR:**
  - Hierarchical tree of summarised chunks
  - Powerful but expensive (many LLM calls)
- **Mix of Granularity:**
  - Router picks chunk granularity per query
  - Requires separate training phase

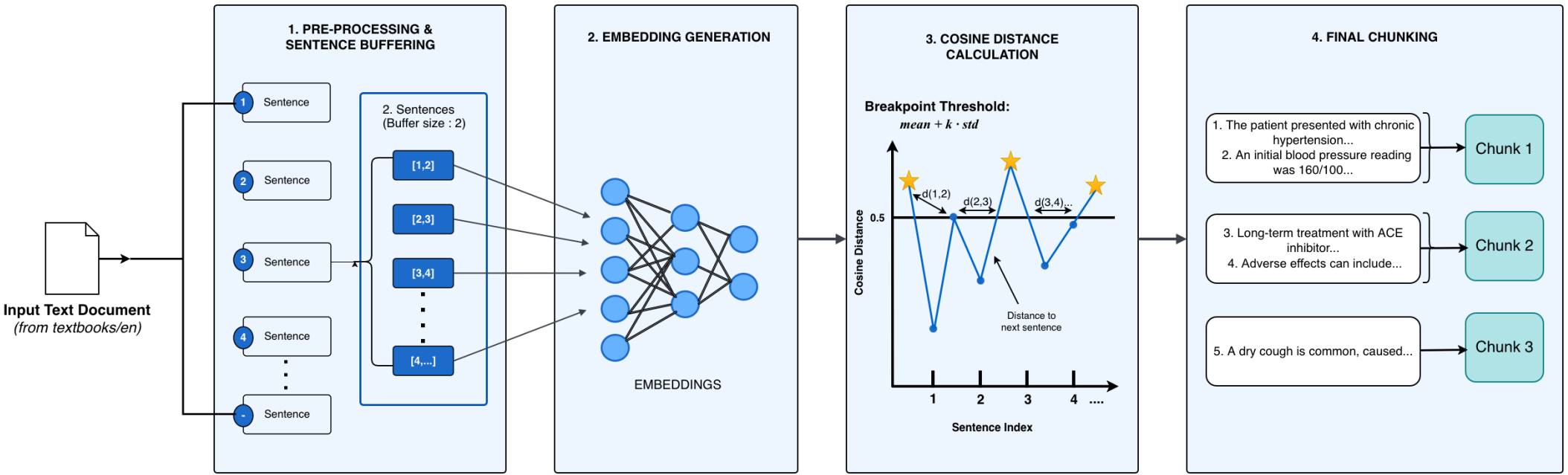


Chunking Strategies

Fixed Size Chunking: A Simple Approach

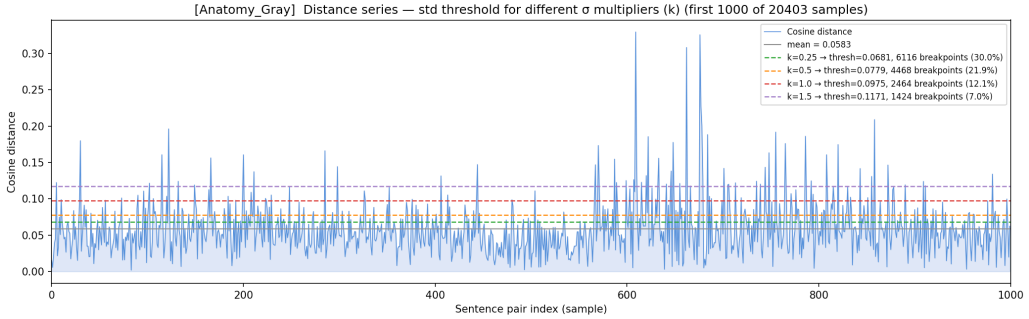
- **Advantage:** Only requires one Hyper-parameter (Chunk Size)
- **Limitations:**
  - **Context Loss:** Loss of context
  - **Fragmentation:** Single sentences can be broken across multiple chunks

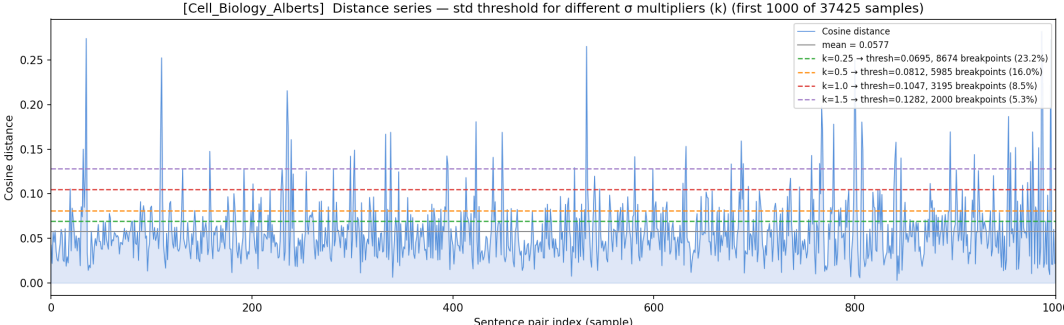
Semantic Chunking



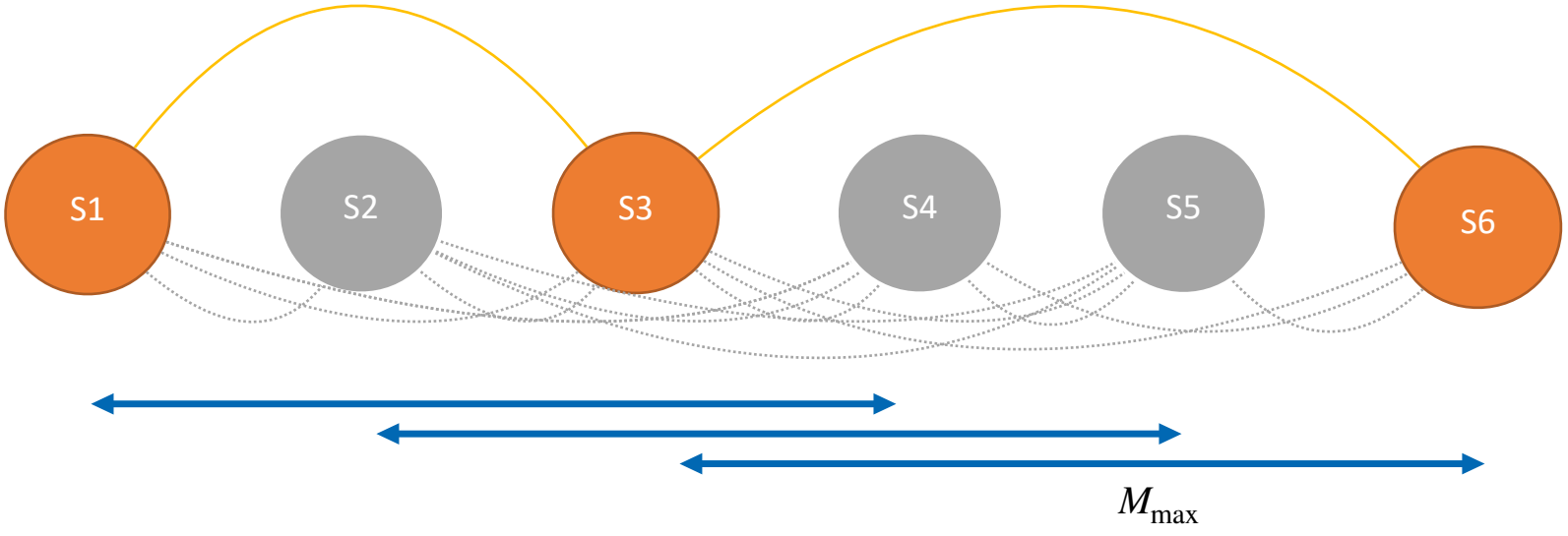
Semantic Chunking Limitations

- **Difficult Hyperparameter Tuning:**
  - Requires balancing buffer sizes and similarity thresholds
- **Context vs Compute Trade-Off:**
  - High buffer size = better context, higher cost
  - Low buffer size = lower cost, fragmented information
- **Threshold Variability:**
  - Optimal threshold values are highly dependent on the specific dataset being used





Adaptive Semantic Chunking Method (ASCM)



ASCM : Edge Cost

Edge Cost: $\phi(i,j) = \alpha\, C_{\text{boundary}}(j) + \lambda\, P_{\text{len}}(i,j) + \mu\, P_{\text{coh}}(i,j)$

$$C_{\text{total}}(j) = \delta\, C_{\text{sem}}(s_{j-1}, s_j) + \beta\, C_{\text{struct}}(s_{j-1}, s_j) + \gamma\, C_{\text{med}}(s_{j-1}, s_j)$$

$$C_{\text{boundary}}(j) = \frac{1}{1 + C_{\text{total}}(j)}$$

$C_{\text{sem}}(s_{j-1}, s_j)$: Cosine similarity

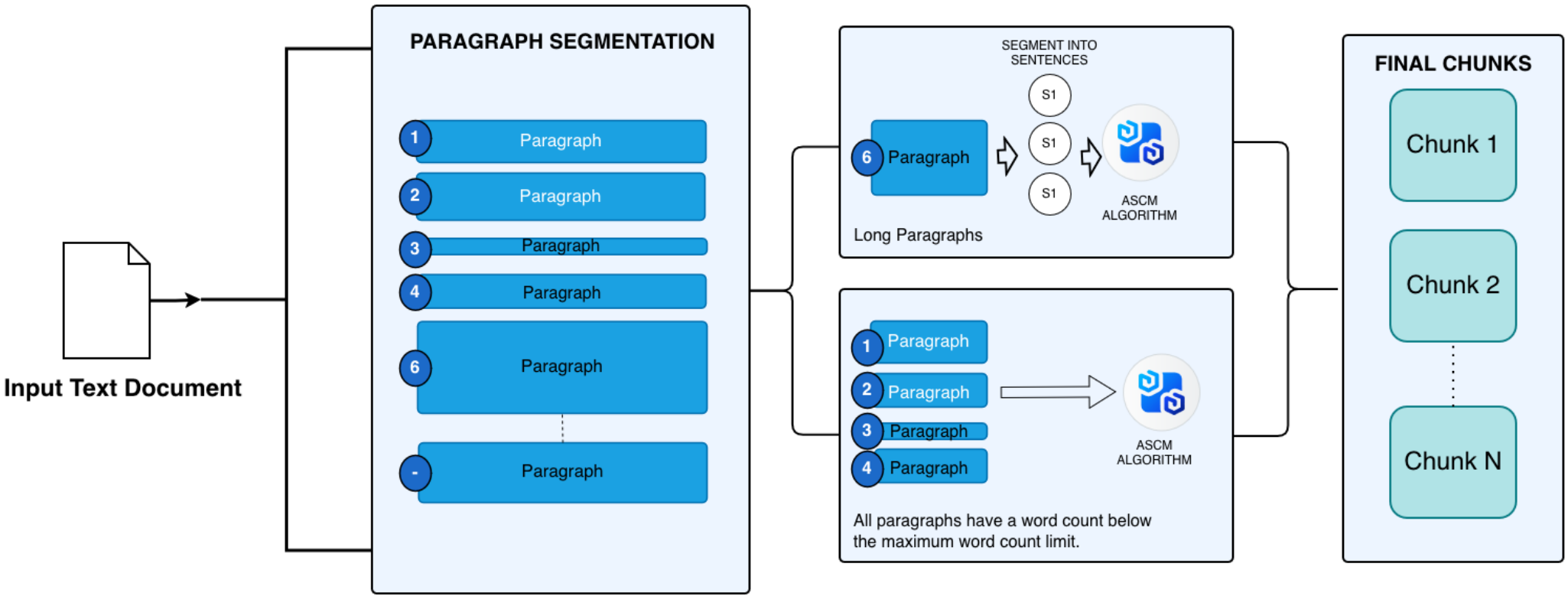
$C_{\text{struct}}(s_{j-1}, s_j)$: Structural relation (**Hard to determine in advance.**)

$C_{\text{med}}(s_{j-1}, s_j)$: Medical constraint (**Hard to determine in advance.**)

$P_{\text{len}}(i,j)$: Penalty for deviation from optimal size

$P_{\text{coh}}(i,j)$: Penalty for low internal connectivity

ASCM : C_{struct} Solution



ASCM : Edge Cost (Reformulated)

$$C_{\text{total}}(j) = \cos\left(\mathbf{s_{j-1}}, \mathbf{s_j}\right)$$

$$L_{\text{opt}}(i) = \text{max sentences with total words} \leq M_{\max}$$

$$P_{\text{len}}(i,j) = \frac{\left(L(i,j) - L_{\text{opt}}(i)\right)^2}{L_{\text{opt}}(i)}$$

$$P_{\text{coh}}(i,j) = \text{average distance between sentences}$$

ASCM Chunking via Dynamic Programming

Algorithm 1 ASCM Chunking via Dynamic Programming

Require: Sentences $S = \{s_0, \ldots, s_{n-1}\}, M_{max}, M_{min}$
Ensure: Optimal segmentation of sentences into chunks

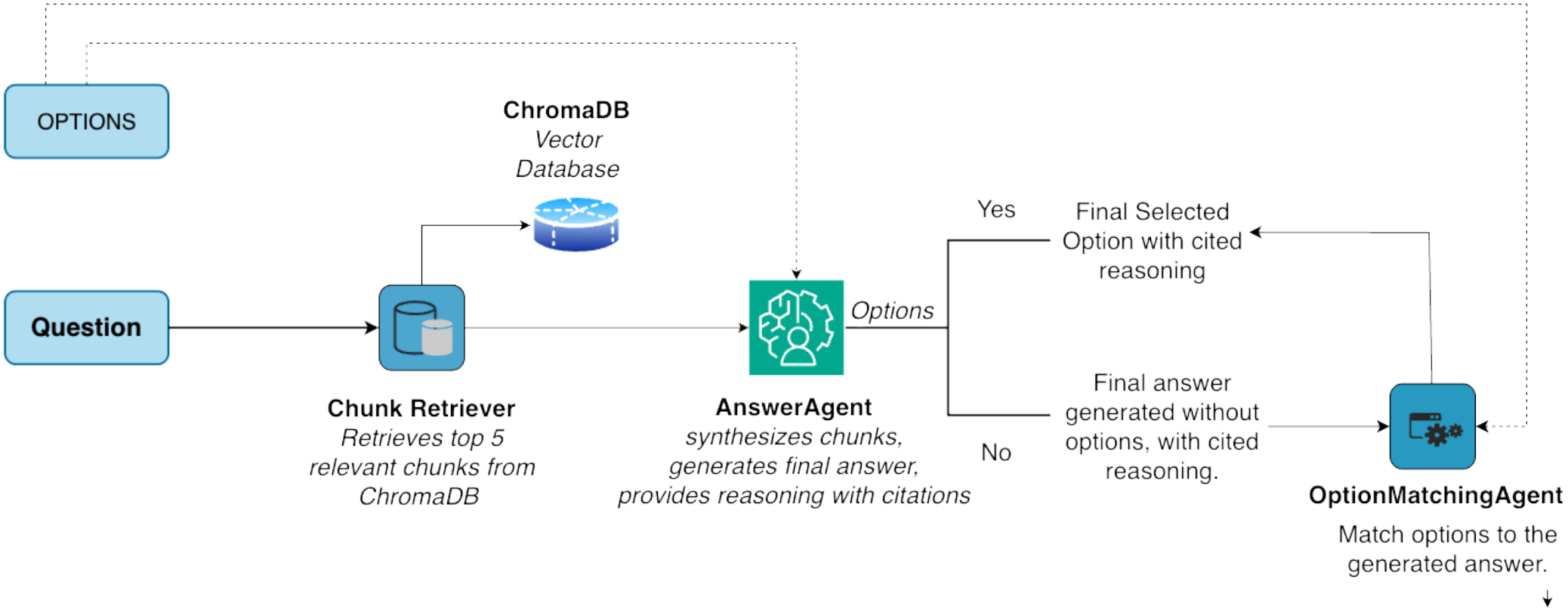
```
n ← |S|
dp[0...n] ← +∞,    dp[0] ← 0
parent[0...n] ← −1
Compute optimal chunk size parameter via L_opt
for j = 1 to n do
    Determine (L_min, L_max) for position j
    for i = max(0, j − L_max) to j − L_min do
        cost ← dp[i] + φ(i, j)
        if cost < dp[j] then
            dp[j] ← cost
            parent[j] ← i
        end if
    end for
end for
return parent {Backtrack to recover optimal chunks} =0
```



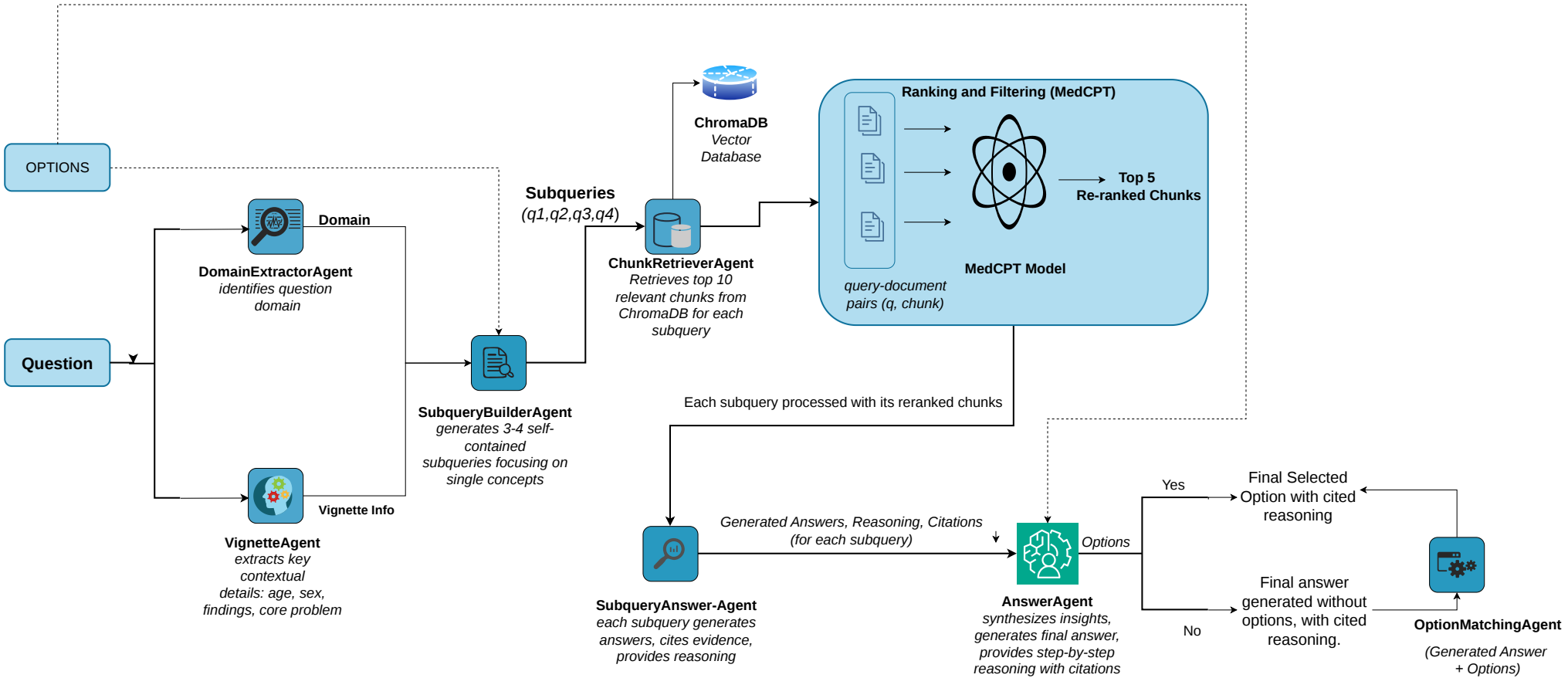


Model Architectures

Simple RAG / Naive RAG



Multi-Agent


OPTIONS
Question
Domain
DomainExtractorAgent
identifies question domain
Vignette Info
VignetteAgent
extracts key contextual details: age, sex, findings, core problem
SubqueryBuilderAgent
generates 3-4 self-contained subqueries focusing on single concepts
Subqueries
(q1,q2,q3,q4)
ChromaDB
Vector Database
ChunkRetrieverAgent
Retrieves top 10 relevant chunks from ChromaDB for each subquery
Ranking and Filtering (MedCPT)
MedCPT Model
Top 5
Re-ranked Chunks
query-document pairs (q, chunk)
Each subquery processed with its reranked chunks
SubqueryAnswer-Agent
each subquery generates answers, cites evidence, provides reasoning
Generated Answers, Reasoning, Citations
(for each subquery)
AnswerAgent
synthesizes insights, generates final answer, provides step-by-step reasoning with citations
Options
Yes
Final Selected Option with cited reasoning
No
Final answer generated without options, with cited reasoning.
OptionMatchingAgent
(Generated Answer + Options)

Experimental Setup

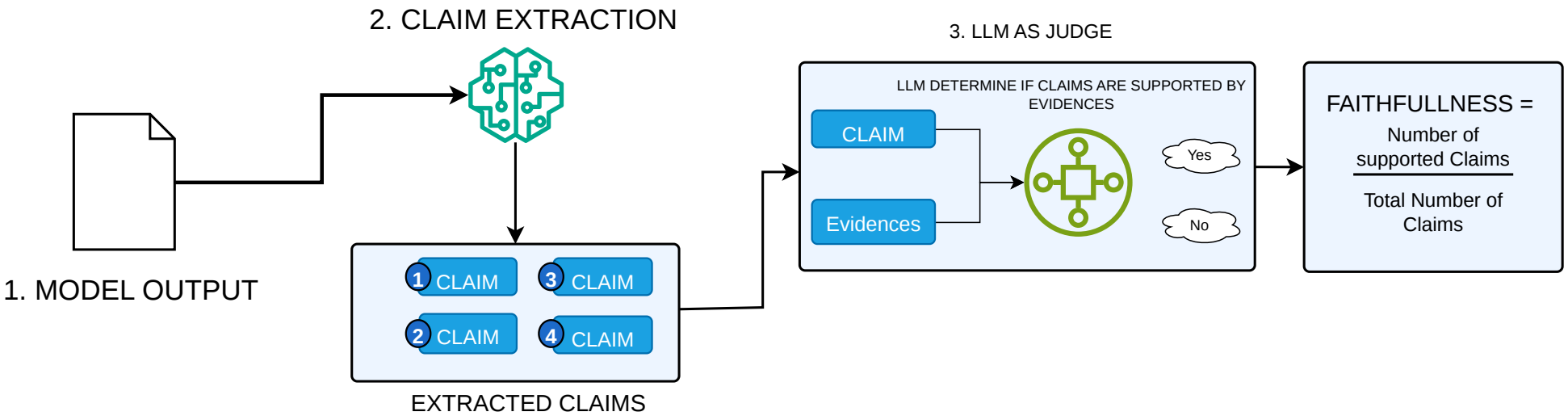
Chunking Methods Comparison

- Fixed Size Chunking
  - Chunk Size : 150
- Semantic Chunking
  - Buffer size : 2
  - Threshold Method : Standard deviation Based (mean + k * std)
- ASCM
  - Max Tokens : 50
  - Min Tokens : 150
- Model Used : LLaMA-3.1-8B-Instruct

Single Agent vs Multi Agent

- Model Used : LLaMA-3.1-8B-Instruct and GPT-OSS-20B
- Both models with Same Prompt
- With and without options inference
  - Analyse whether model relies on Patter matching
  - Examine whether the model can predict correct answer independently

Faithfulness Evaluation



Results

Chunking Method Performance

Method	Chunk Used	Accuracy (%)	Correct / Total
FixedSize	5	56.48	719 / 1273
Semantic Chunking	5	56.09	714 / 1273
ASCM	5	**57.42**	**731 / 1273**

Table 1. Comparison of chunking methods. Best results are shown in bold.

Single-Agent vs. Multi-Agent (with Options): Results

Setup	Model	Accuracy (%)	Correct / Total
SingleAgent	openai/gpt-oss-20b	63.08	803 / 1273
	llama-3.1-8b-instant	58.6	746 / 1273
Multiagent	llama-3.1-8b-instant	57.8	736 / 1273
	openai/gpt-oss-20b	**73.2**	**932 / 1273**

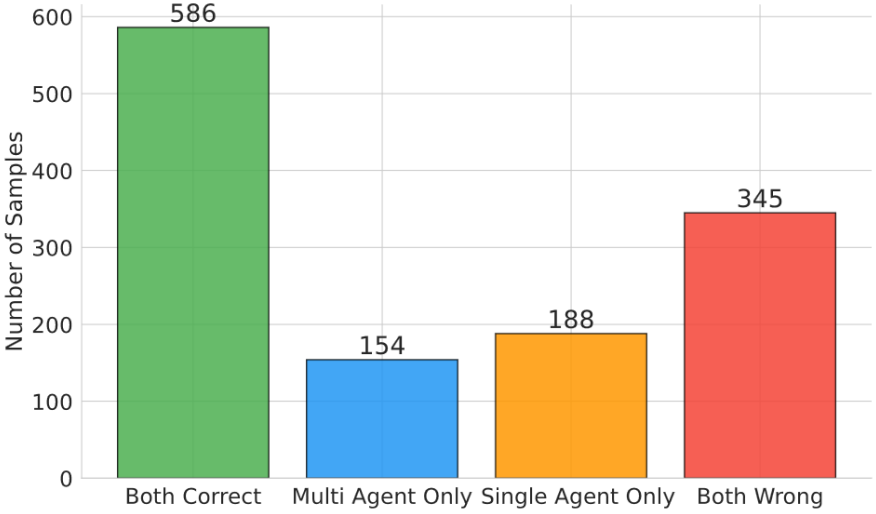
Table 2. Performance comparison between SingleAgent and Multiagent models. Best results are in bold.

Single-Agent vs. Multi-Agent (without Options): Results

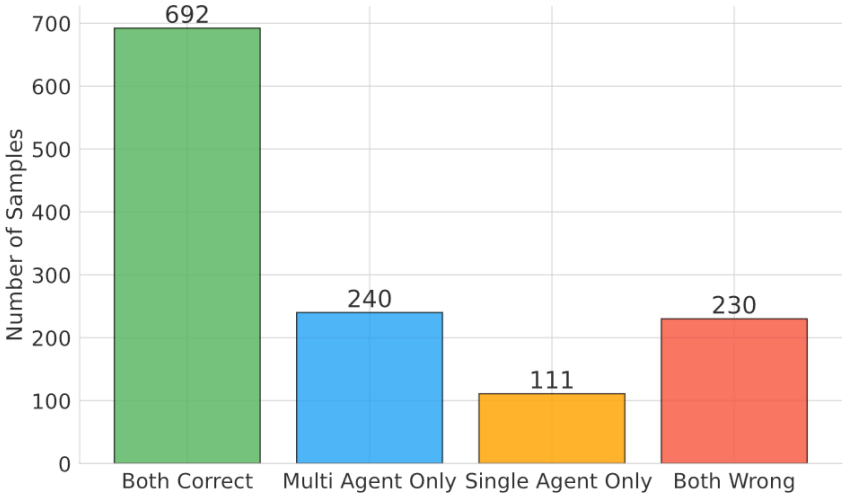
Setup	Model	Accuracy (%)	Correct / Total
SingleAgent	openai/gpt-oss-20b	60.8	774 / 1273
	llama-3.1-8b-instant	43.36	552 / 1273
Multiagent	openai/gpt-oss-20b	58.1	740 / 1273
	llama-3.1-8b-instant	46.1	587 / 1273

Table 3. Performance comparison of SingleAgent and Multiagent models in open-ended (no options) setting.

Options Unlock Multi-Agent Performance

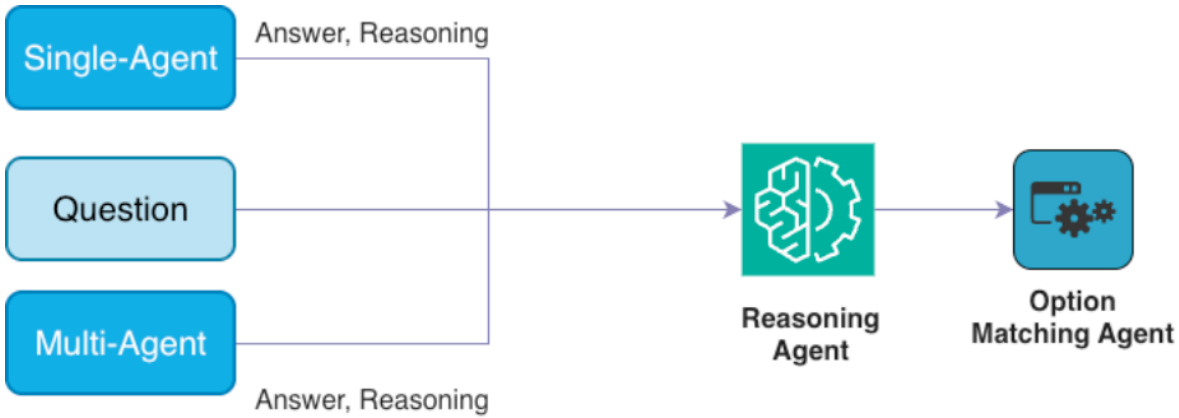


Without Options



With Options

Performance Gains from Reasoning-Guided Ensemble

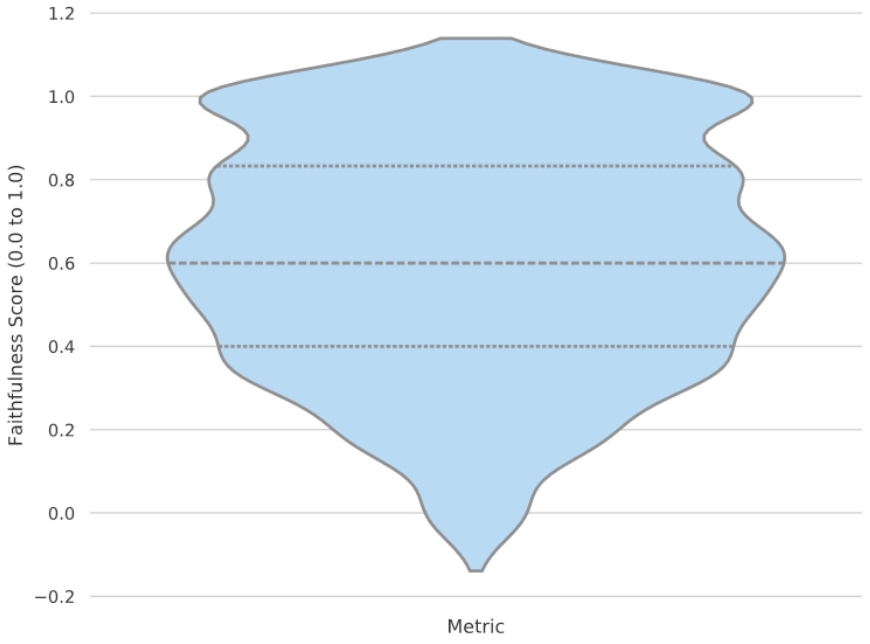


Setup	Model	Accuracy (%)	Correct / Total
SingleAgent	openai/gpt-oss-20b	60.8	774 / 1273
	llama-3.1-8b-instant	43.36	552 / 1273
Multiagent	openai/gpt-oss-20b	58.1	740 / 1273
	llama-3.1-8b-instant	46.1	587 / 1273
Ensemble	openai/gpt-oss-20b	**69.13**	**880 / 1273**

Table 3. Performance comparison of SingleAgent and Multiagent models in open-ended (no options) setting. Best results are shown in bold.

Faithfulness Remains a Bottleneck

- ~ 0.60 of generated claims are supported by retrieved evidence
- Majority of samples fall between 0.4-0.8
- Small subset with near-zero faithfulness
- Even with strict prompting LLM hallucinated




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Human Centered Artificial Intelligence

Summary

Summary

- Two key dimensions explored :
  - Document chunking
  - Multi agent architecture
- Chunking insights :
  - ASCM outperformed fixed size and semantic chunking
- Key observations:
  - Improves performance only with strong reasoning model
  - Subquery decomposition not always helpful
  - Single agent and multi agent system shows complementary strengths
  - Ensemble approach achieved best performance (No Options)
- Faithfulness challenge:
  - Median Score ~0.6 : Even though with strict prompting model use internal knowledge



Thank you!
Questions?